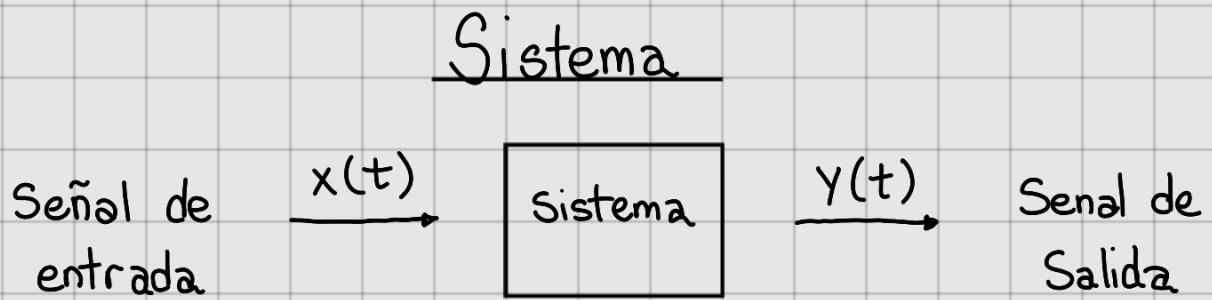




- * Una variable independiente $\Rightarrow$ Tiempo
 - Continuo: $[t]$
 - Discreto: $[n]$
- * Una variable dependiente 'y'

En tiempo continuo la señal existe en cada momento de t
 En tiempo discreto la señal existe en los enteros ' n '

Ambas se analizan desde ∞^- hacia ∞^+ .

Transformaciones que puede sufrir la señal son:

- Desplazamiento en el tiempo: Sumar o restar un número

$$x(t - t_0) \quad \text{ó} \quad x(t + t_0)$$

- Escalamiento: se multiplica por un número.

$$x(t \cdot a) \quad \text{ó} \quad x(t/a)$$

- Reflexión: Se multiplica por -1 .

$$x(-t)$$

Prioridad de combinación de transformaciones

- 1: Desplazo
- 2: Escalo
- 3: Reflejo

Señales básicas: Señales que nos permiten representar señales más complejas.

- Impulso unitario: Su área es uno (1).

$$\delta(t) \begin{cases} \infty & \text{para } t = 0 \\ 0 & \text{para } t \neq 0 \end{cases} \quad \int_{-\infty}^{\infty} \delta(t) \cdot dt = 1$$

* Se le puede aplicar desplazamiento pero el escalado y la reflexión no tienen sentido, no le aplica ningún cambio.

Representación Gráfica



Representación gráfica incremental

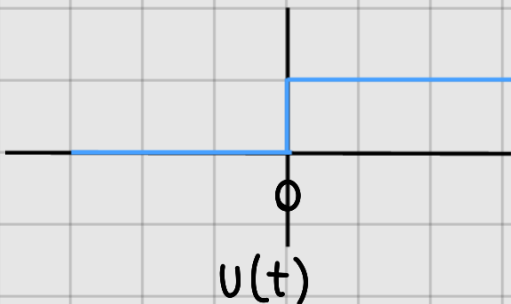


$$\delta_{\Delta}(t) = \begin{cases} 0 < t < \Delta \rightarrow \delta_{\Delta}(t) = \frac{1}{\Delta} \\ \text{otro } t \rightarrow \delta_{\Delta}(t) = 0 \end{cases}$$

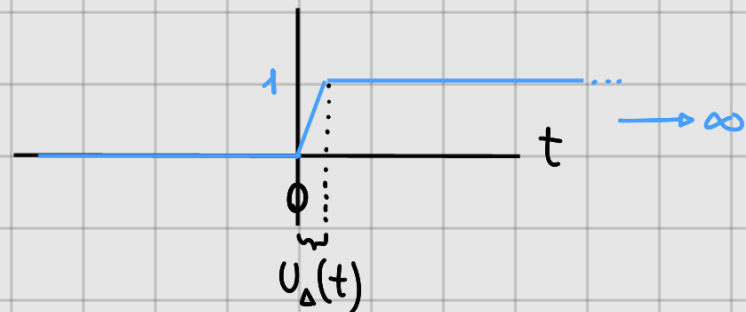
• Escalon unitario : Su area es ∞ desde $t_0 = 0$ con valor de señal igual a uno (1)

$$U(t) = \begin{cases} t \geq 0 \rightarrow U(t) = 1 \\ t < 0 \rightarrow U(t) = 0 \end{cases}$$

Representación Gráfica



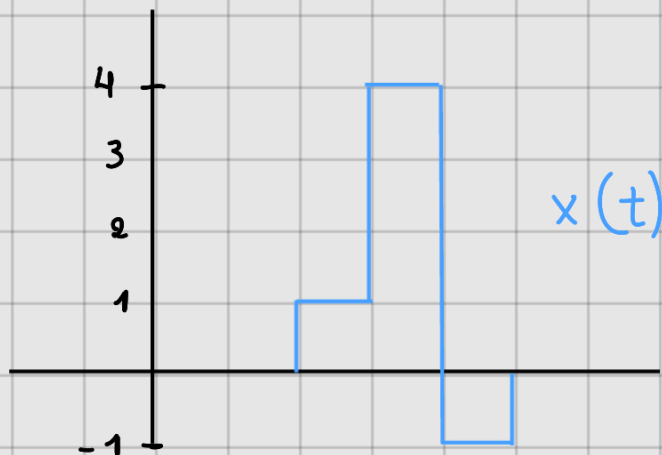
Representación gráfica incremental



$$\lim_{\Delta \rightarrow 0} U_{\Delta}(t) = U(t)$$

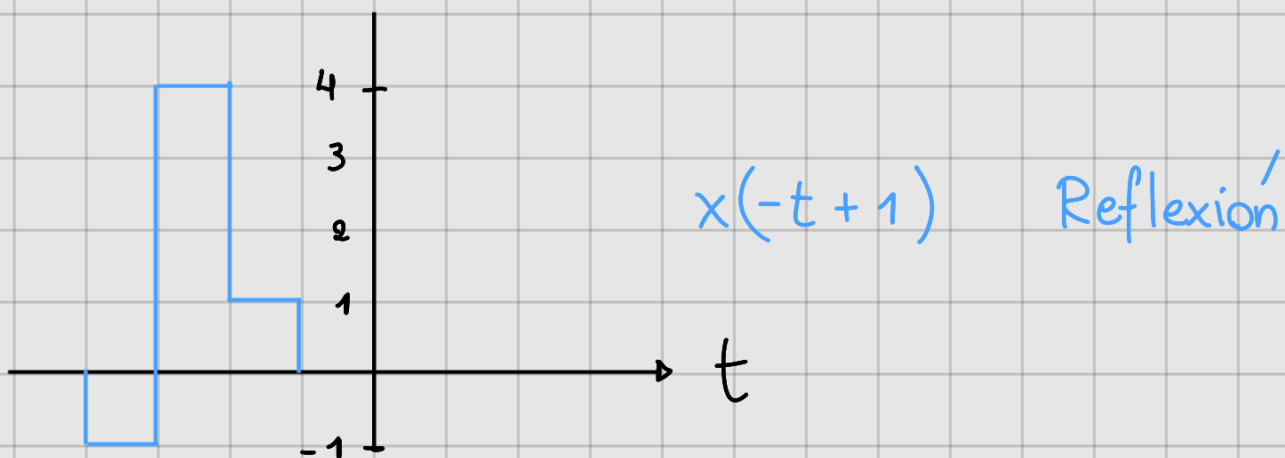
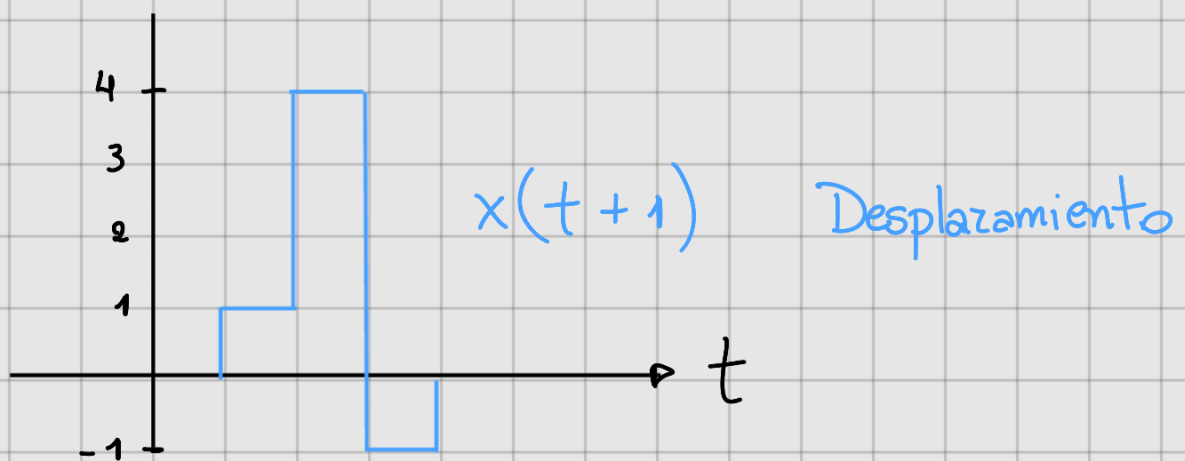
Ejercicio

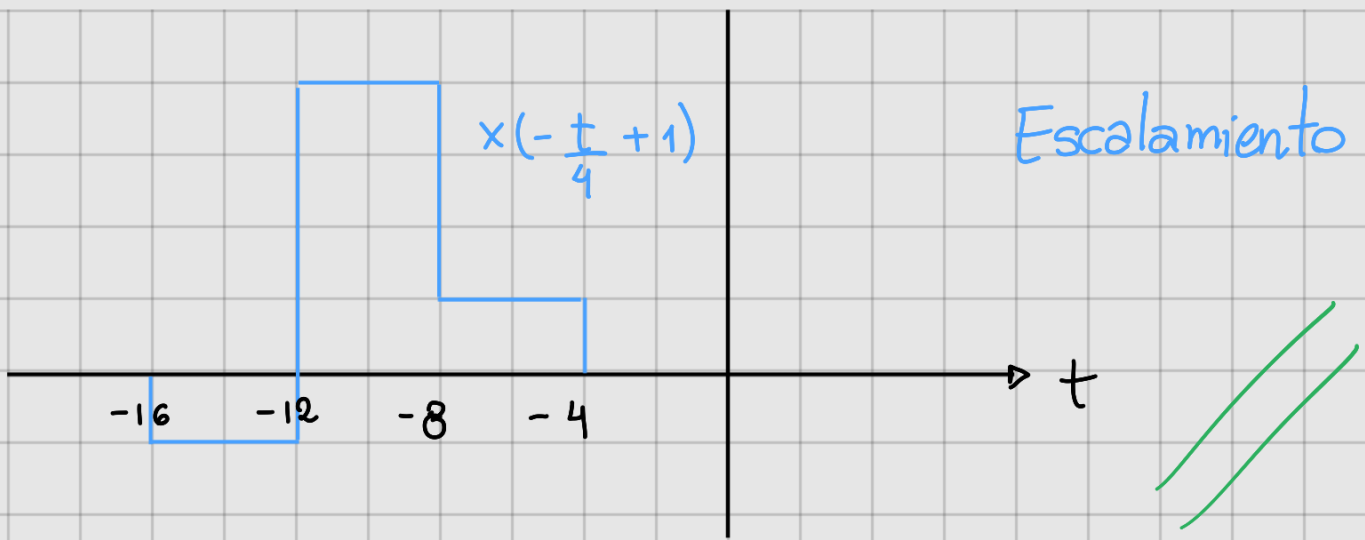
Dado:



Obtener $x\left(-\frac{t}{4} + 1\right)$

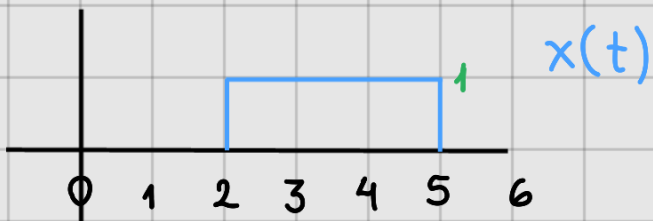
Solución:





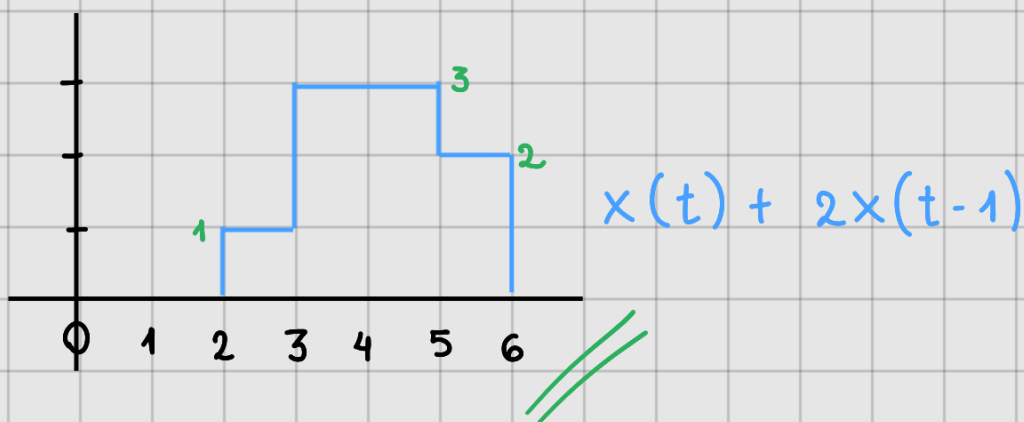
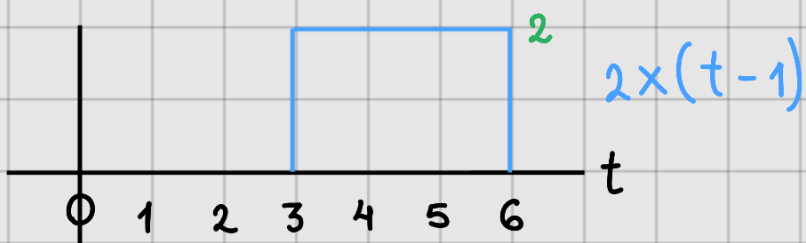
Ejercicio 2

Dado

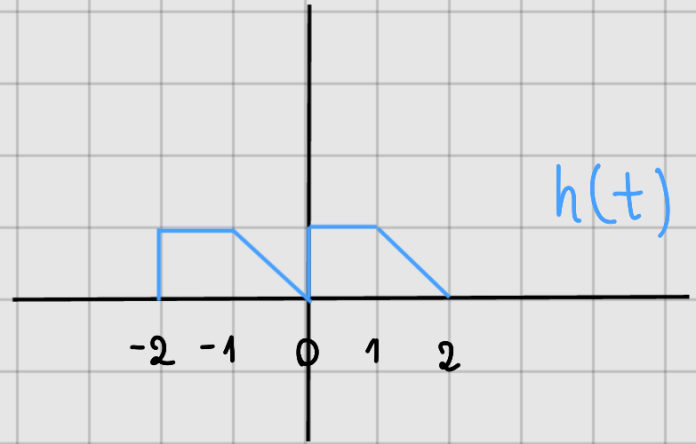
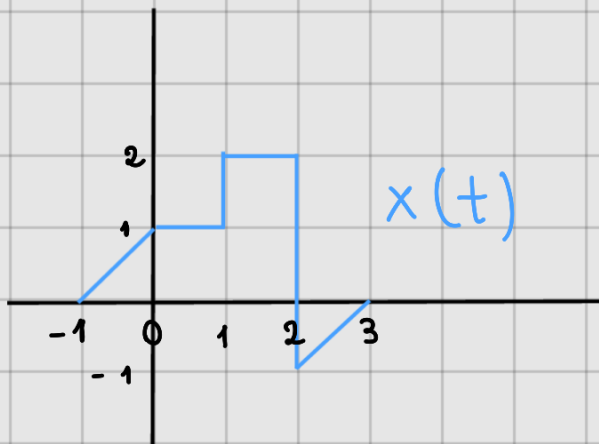


Obtener: $x(t) + 2x(t-1)$

Solucion:

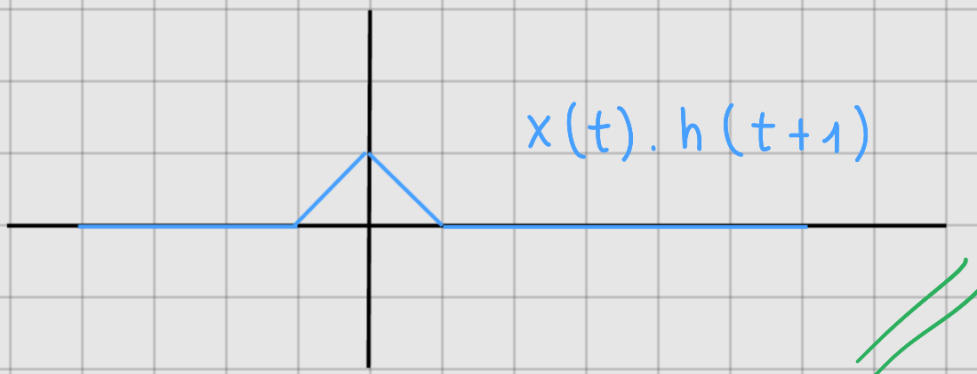
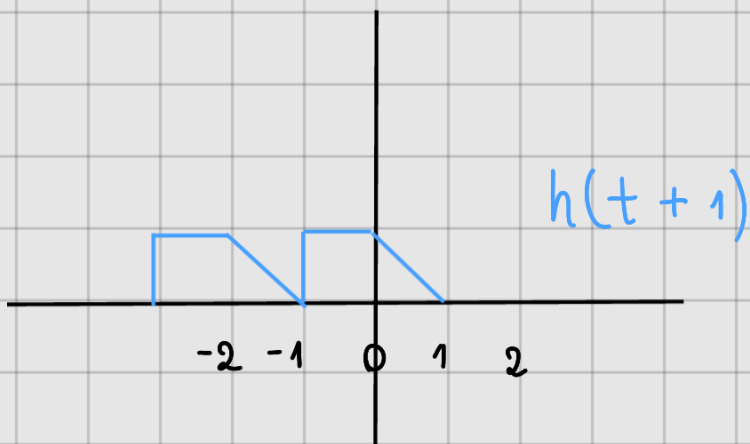


Ejercicio 2.9)



Se pide: $x(t) \cdot h(t+1)$

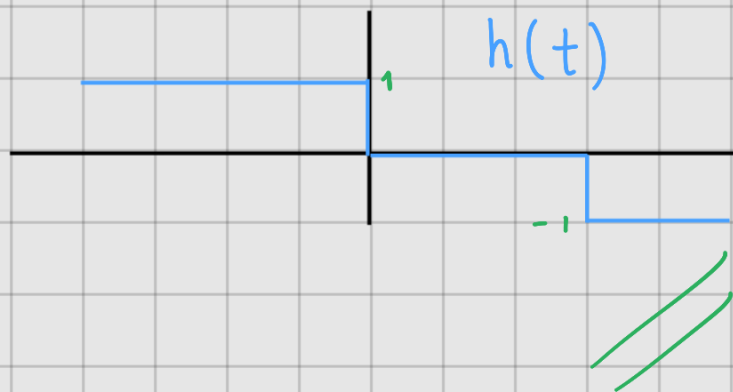
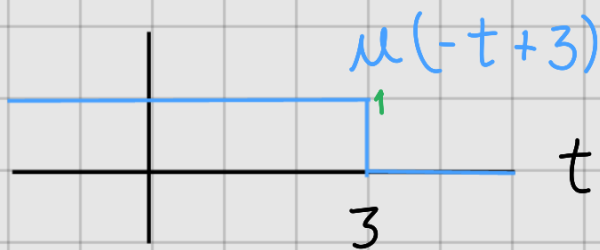
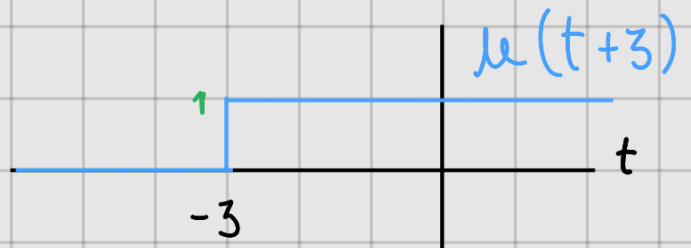
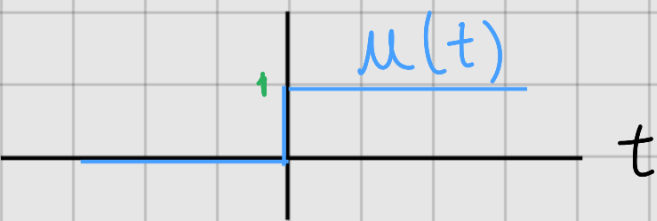
Solución



Ejercicio Escalón

$$h(t) = \mu(-t+3) - \mu(t)$$

Solución

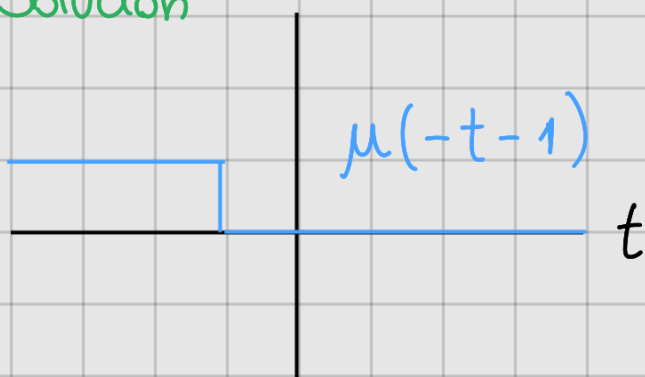


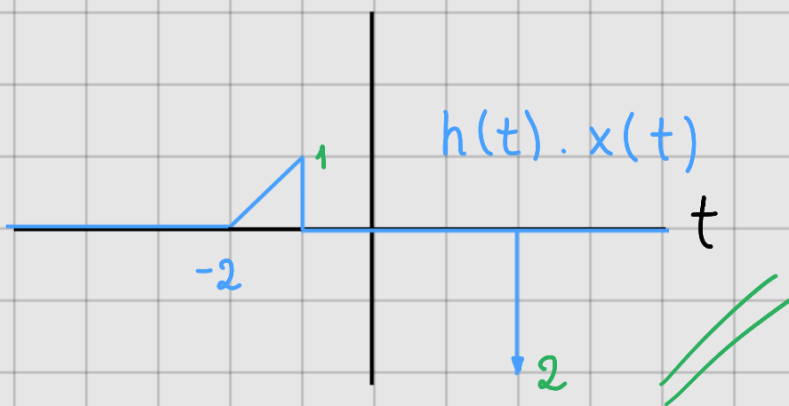
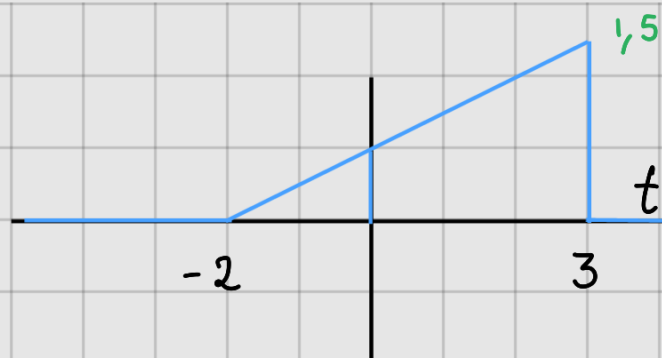
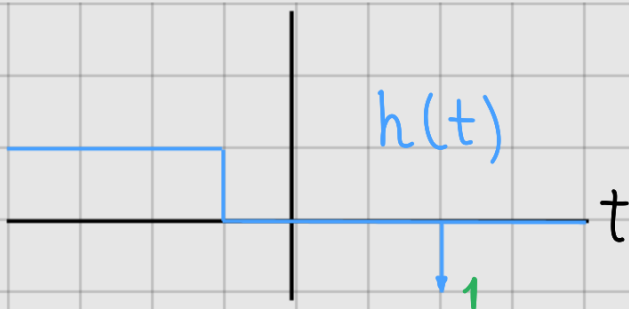
Ejercicio Impulso

$$h(t) = \mu(-t-1) - \delta(t-2) \quad x(t) = \begin{cases} t/2 + 1 & -2 < t < 3 \\ 0 & \text{otro } t \end{cases}$$

Se pide $h(t) \cdot x(t)$

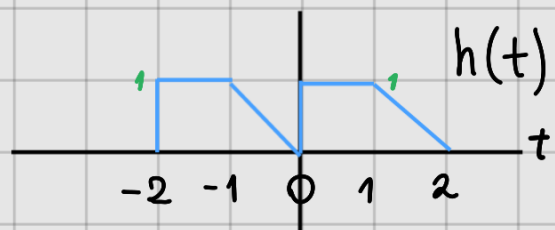
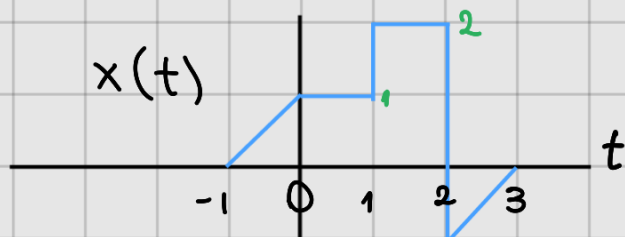
Solución





Ejercicios Propuestos

Dados :



Calcular

I) $4 \cdot h(t/4)$

II) $\frac{1}{2} \cdot h(t) \cdot \mu(t) + h(-t) \cdot \mu(t)$

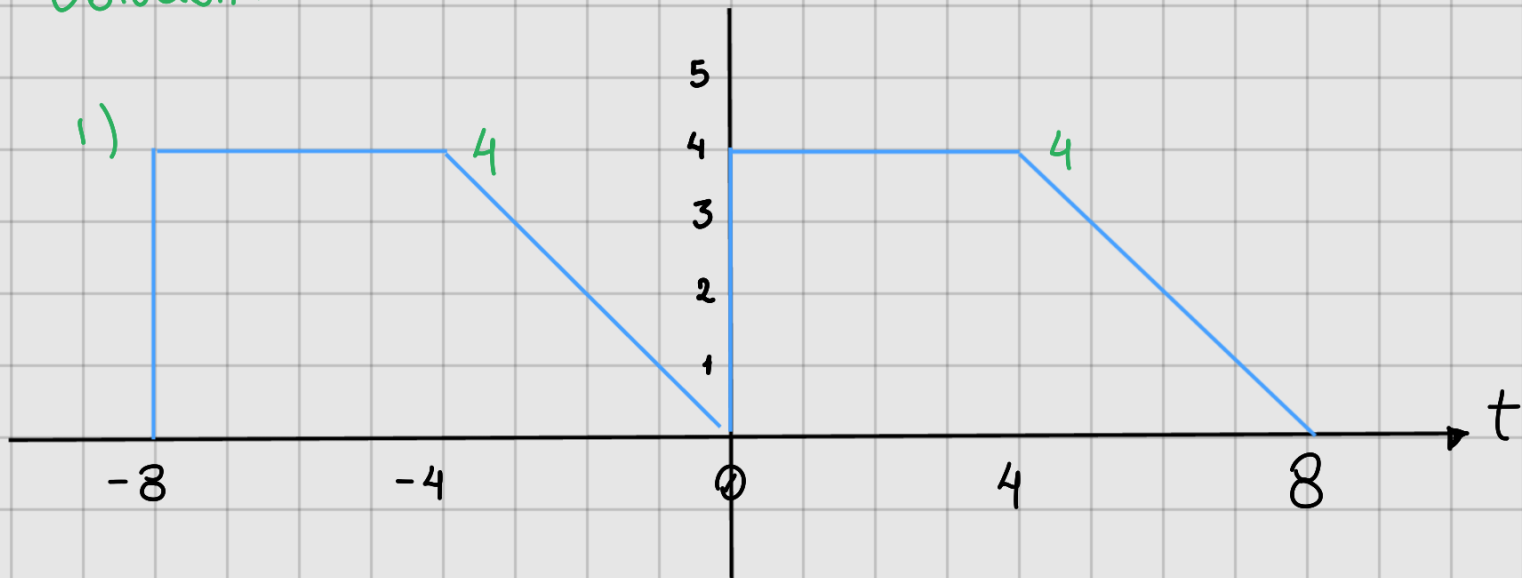
III) $h(t/2) \cdot \delta(t+1)$

IV) $h(t) \cdot [\mu(t+1) - \mu(t-1)]$

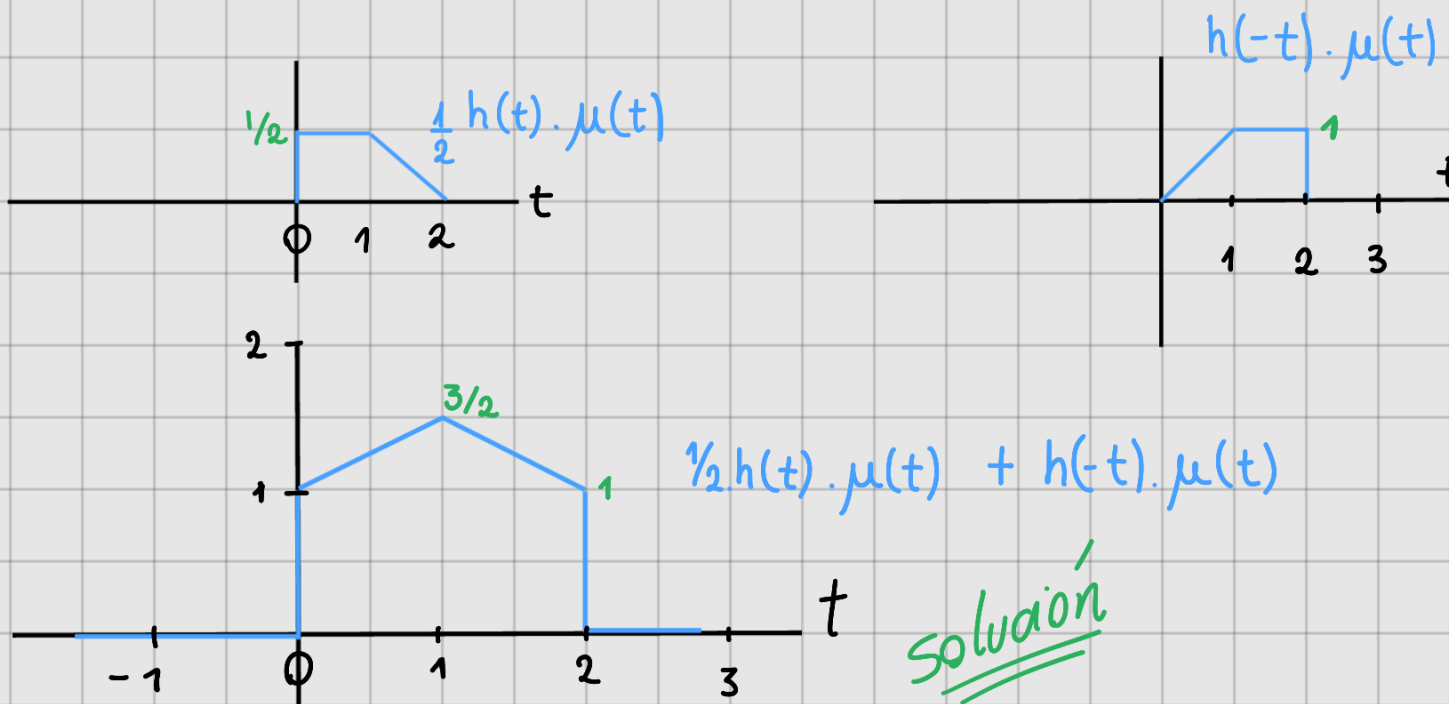
V) $x(2 - t/2) \cdot h(t+4)$

VI) $x(1 - t) \cdot h(t-1)$

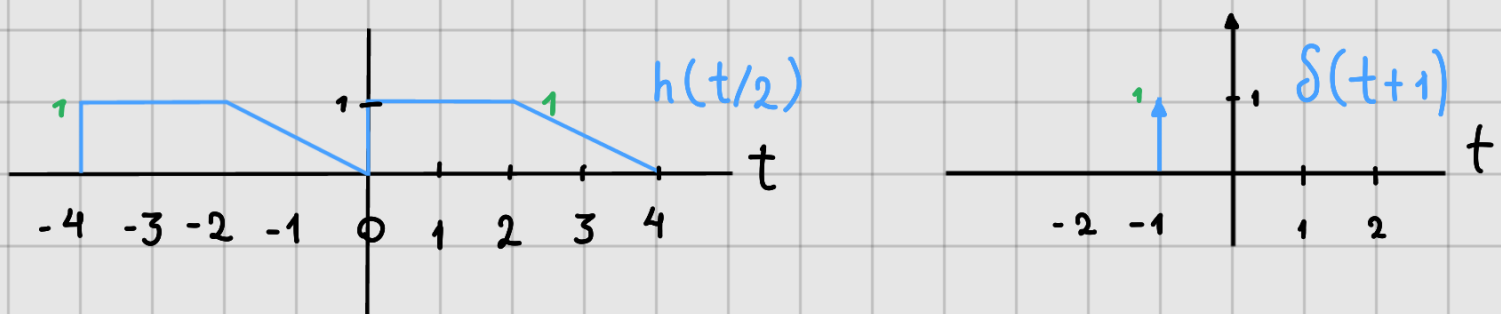
Solución:

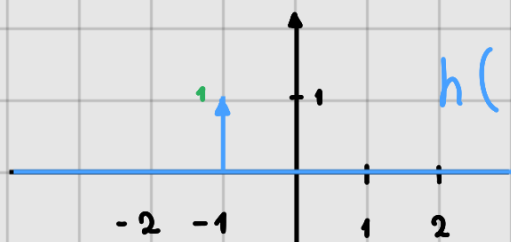


ii)



iii)

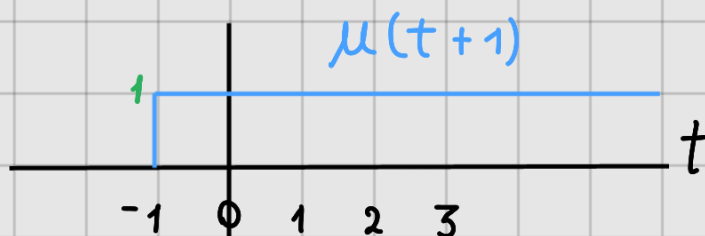




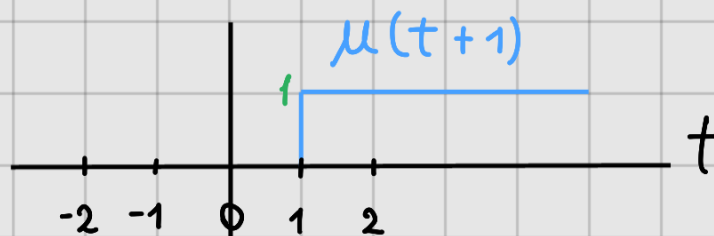
$$h(t/2) \cdot \delta(t+1)$$

Solución

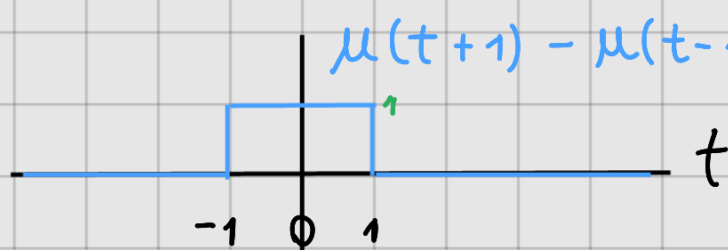
IV)



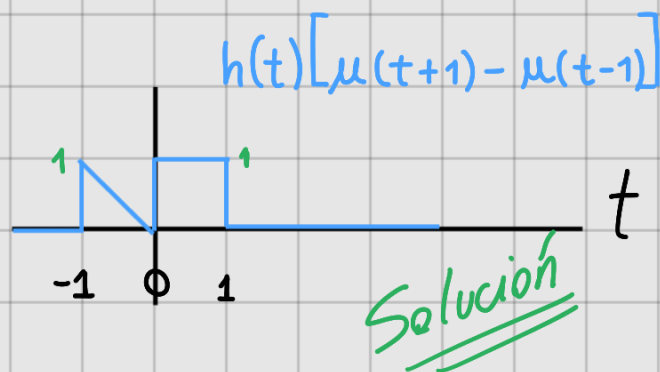
$$\mu(t+1)$$



$$\mu(t+1)$$



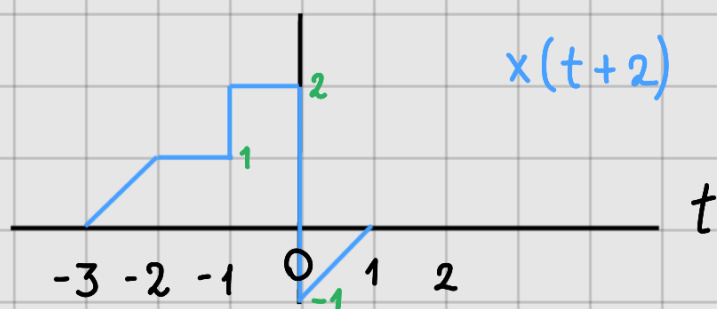
$$\mu(t+1) - \mu(t-1)$$



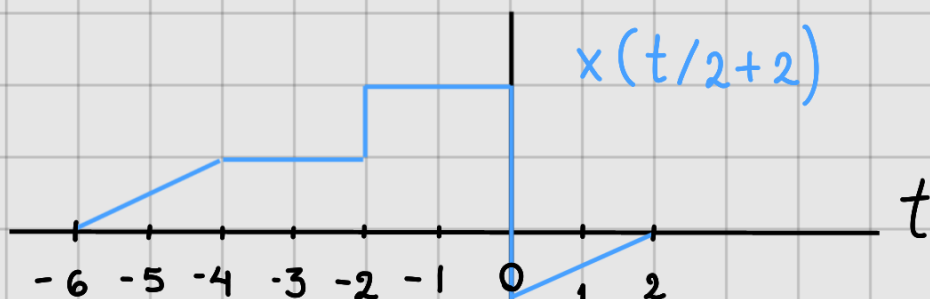
$$h(t)[\mu(t+1) - \mu(t-1)]$$

Solución

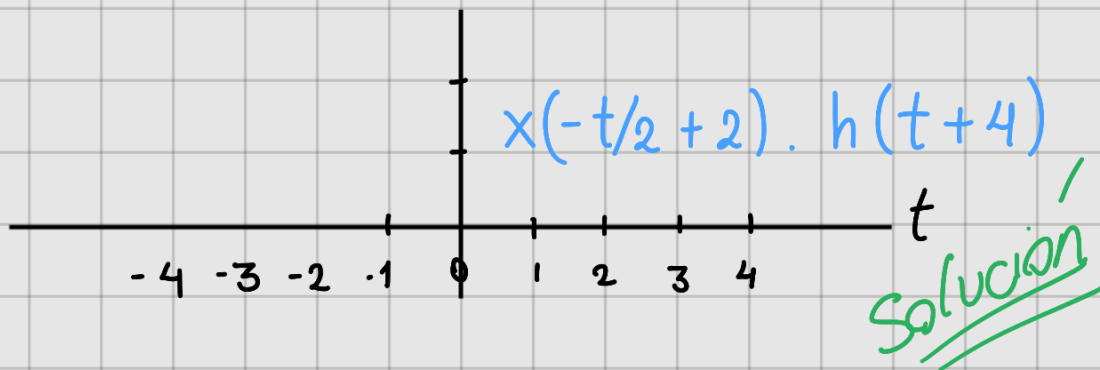
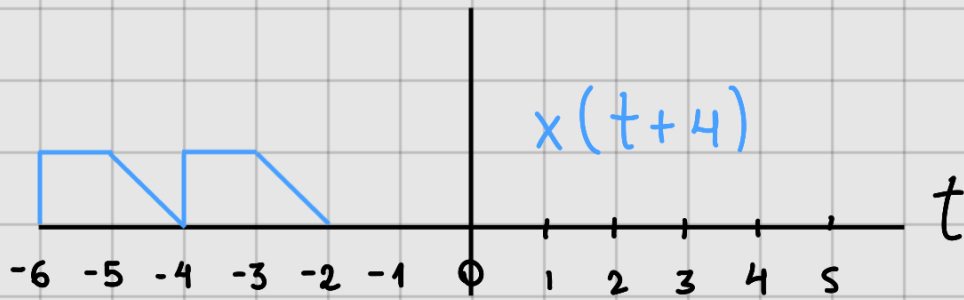
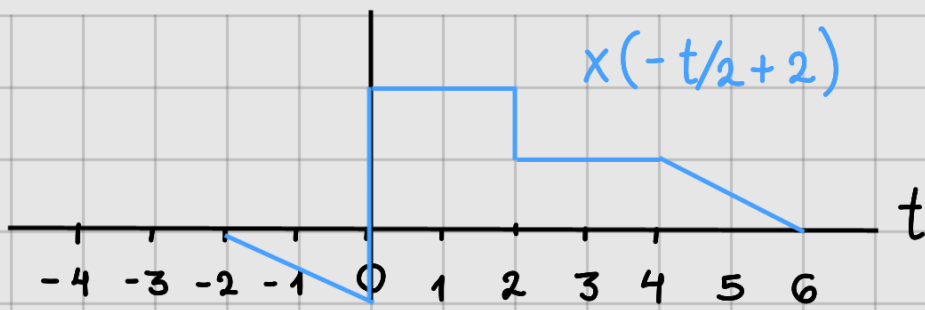
V)



$$x(t+2)$$



$$x(t/2+2)$$



v1)

